American Computer Science League

2024 -2025 ● Contest 2: Short Problems Solutions ● Intermediate Division

1. Prefix/Infix/Postfix Notation

+ − * 2 3 / 4 2 / * 8 + 1 3 4
= + − (* 2 3)(/ 4 2) / * 8 (+ 1 3) 4
= + (− 6 2) / (* 8 4) 4
= + 4 (/ 32 4)
= + 4 8
= 12

E. 12

2. Prefix/Infix/Postfix Notation

$$\frac{xy + z}{a^2}$$

= (x*y + z) / (a^2)
= (x y * z +) / (a^2)
= (x y * z +) / (a 2 ^)
= (x y * z +) (a 2 ^) /
= x y * z + a 2 ^ /

C. x y * z + a 2 ^ /

3. Bit-String Flicking

10110 OR 11011 AND 00101 XOR 01101
= 10110 OR ((11011 AND 00101) XOR 01101)
= 10110 OR (00011 XOR 01101)
= 10110 OR 01110
= 11110

A. 11110

4. Bit-String Flicking Let X = abcde and NOT X = ABCDE LHS = (NOT (RCIRC-3 (LSHIFT-2 X))) = (NOT (RCIRC-3 (LSHIFT-2 abcde))) = (NOT (RCIRC-3 cde00)) = NOT e00cd = E11CD RHS = 01100 LHS = RHS ⇒ E11CD = 01100 ⇒ a = *, b = *, C = 0, D = 0, E = 0 Therefore X = **111	D. **111
5. LISP (ADD 2 (SUB 7 3) 3 (MULT 2 5) 8) = (ADD 2 7÷3 3 2*5 8) = (ADD 2 4 3 10 8) = 2 + 4 + 3 + 10 + 8 = 27	E. 27
6. LISP (CDR (CAR (CDR (CDR '(e (f g) (d b (a c))))))) = (CDR (CAR (CDR '((f g) (d b (a c)))))) = (CDR (CAR '((d b (a c))))) = (CDR '(d b (a c))) = (b (a c))	B. (b (a c))